# Module II – AC Circuits

## Assignment Questions

1. An alternating voltage is given by V(t)=230 sin 314t. Calculate i) Frequency, ii) Maximum value of voltage, iii) Average value of voltage, iv) RMS value of voltage, v) Form Factor and vi) Peak Factor.
2. In a series RL circuit, the current and voltage are i(t) = 5 sin (314t + 3π) and v(t) =20 sin (314t+2π). Determine Impedance of the circuit, Resistance, Inductance, Power factor. Also find power drawn by the circuit?
3. Calculate i) maximum value, ii) average value, iii) RMS value, vi) Form Factor, v)Peak Factor vii) Peak-Peak Current value of the given waveform in Figure 1

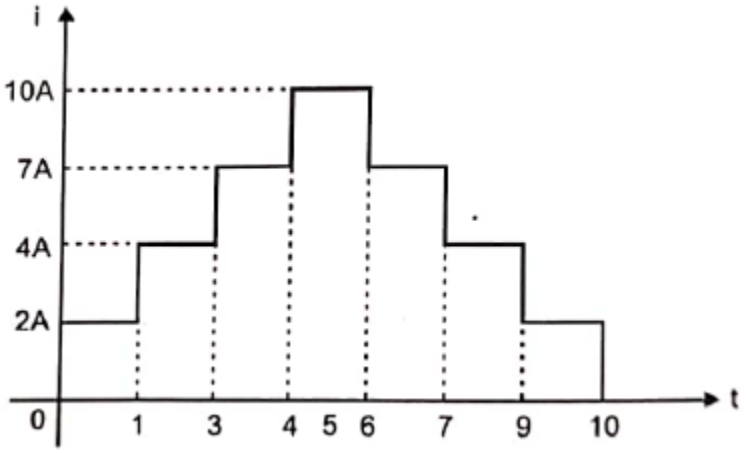


Figure 1

4. A coil of resistance 15 Ω and inductance 130 mH is connected in series with a 220 μF capacitor across a 200 V, 50Hz supply. Estimate the impedance of the circuit, current, power factor, total power, real power, reactive power, voltage across the coil and the capacitor.
5. A capacitor having a capacitance of 10 μF is connected in series with a non-inductive resistance of 120 Ω across 100 V, 50 Hz supply. Find circuit impedance, current, real power, reactive power and apparent power.
6. A series circuit consisting of a 10Ω resistor, a 100μF capacitor and a 10 mH inductor is driven by a50 Hz AC. voltage source of maximum value 100 volts. Calculate the equivalent impedance, current in the circuit, inductive reactance, capacitive reactance and the phase angle.

7. An alternating current is given by i(t) = 25 sin 75t. Calculate i) Frequency, ii) Maximum value of current, iii) Average value, iv) RMS value, v) Form Factor, vi) Peak Factor.
8. An inductive load takes 10 A from a 230 V, 50 Hz single-phase supply at a power factor of 0.8 lagging.
   (a) Calculate the real power (kW).
   (b) Calculate the reactive power (kVAR).
   (c) Find the apparent power (kVA).
9. A coil having a resistance 15 Ω and capacitance of 0.5 F is connected across 220 V, 50 Hz supply as shown in Figure 2. Calculate the inductive capacitance, impedance, current, power factor, active power and reactive power.

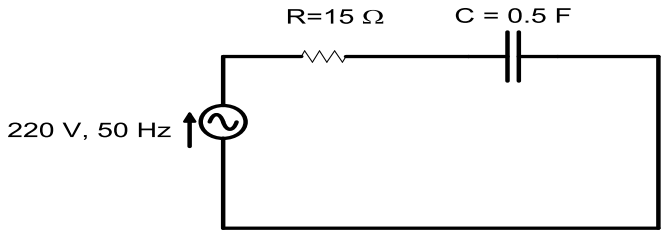


Figure 2

10. In a series RLC circuit, R = 10 Ω, L = 12 mH, C = 20 μF are connected to 220 V, 50 Hz supply. Calculate inductive reactance, capacitive reactance, impedance, current, phase angle, power factor, real power and apparent power.
11. An AC source of 230 V, 50 Hz is applied to a series circuit containing a resistor of 10 Ω, an inductor of 0.2 H and a capacitor of 50 μF. Determine the resonant frequency, current at resonance, bandwidth, half-power frequencies and quality factor (Q).
12. Derive an expression for average value and RMS value of the sine wave.
13. For a circuit consisting of pure Inductance (L) excited by a sinusoidally varying voltage. Derive current, inductive reactance, impedance, phase angle and power factor.
14. For a circuit consisting of pure capacitance (C) excited by a sinusoidally varying voltage. Derive current, inductive reactance, impedance, phase angle and power factor.